📊 Data Analytics Project Activity: From Raw Data to Interactive Dashboard

🎯 Objective

In this activity, you will take a real-world dataset, clean and prepare it (data wrangling), perform exploratory data analysis (EDA), and finally build an interactive dashboard using Python.

This mirrors a real job workflow used by Data Analysts and Business Intelligence professionals.

🧩 Step 1: Dataset Selection (Real-World Data)

📌 Task

Choose **one dataset** from a public source.

Recommended Sources:

- Kaggle: https://www.kaggle.com/datasets
- Google Dataset Search: https://datasetsearch.research.google.com
- Data.gov (Government datasets)
- World Bank Data

📊 Suggested Dataset Types (Pick ONE):

- Retail sales data
- E-commerce transactions
- Covid-19 cases or health data
- Airbnb listings
- Financial stock data
- Sports performance data
- Customer churn data

Requirement

Your dataset must have:

- At least **100+ rows**
- At least **4 columns**
- A time-based column (date/month/year) preferred

Step 2: Data Wrangling (Cleaning & Preparation)

Task

Load your dataset using Python (Pandas) and clean it.

You must perform:

1. Load dataset

```
import pandas as pd
df = pd.read_csv("your_file.csv")
```

2. Handle missing values

- Identify missing values
- Fill or remove them

3. Fix data types

- Convert dates using `pd.to_datetime()`
- Ensure numeric columns are correct

4. Remove duplicates (if any)

5. Create new columns (feature engineering)

Examples:

- Year, Month, Day from Date
- Profit margin = Profit / Sales
- Age groups (if demographic data)

Output Expected:

A **clean dataset ready for analysis**

📊 Step 3: Exploratory Data Analysis (EDA)

📌 Task

Explore the dataset and answer business questions.

🔍 Required Analysis:

1. Basic statistics

- Mean, median, min, max

2. Group analysis

- Sales by category
- Profit by region
- Monthly trends

3. Visualizations (Minimum 4 required)

Use Plotly or Matplotlib:

- Bar chart → Category vs Sales
- Line chart → Trend over time
- Pie chart → Distribution
- Box plot → Spread of values

🧠 Questions to Answer:

- What category generates the highest revenue?
- What time period shows peak performance?
- Are there any unusual patterns or outliers?
- Which region performs best?

✅ Output Expected:

EDA insights + visual charts

Step 4: Build Interactive Dashboard (Final Project)

Task

Create an interactive dashboard using:

Dash (Plotly)

Dashboard Must Include:

1. Filters

- Year dropdown OR category filter

2. Visualizations (minimum 4–5):

- Bar chart (Category vs Sales)
- Line chart (Time trend)
- Pie chart (Distribution)
- Box plot (Spread)
- Optional: Sunburst / Heatmap

Dashboard Features:

- Interactive filtering
- Dynamic charts
- Clean layout
- Clear titles and labels

Bonus (Optional but recommended):

- KPI cards (Total Sales, Profit, Avg Value)
- YoY comparison
- Export dataset feature

Final Submission Requirements

You must submit:

1. Python notebook or script

- Data cleaning + EDA + dashboard code

2. Dataset used

- Original dataset file or link

3. Short report (1–2 pages)

Include:

- Dataset chosen
- Key insights
- Challenges faced
- What you learned

🚀 Learning Outcome

By completing this project, you will understand:

- Real-world data wrangling
- Business analysis thinking
- Data storytelling
- Building dashboards like industry analysts
